

Chapter 27. Transport of the Ventilator-Supported Patient

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Transport of the Ventilator-Supported Patient: Introduction

Mechanical ventilation of the critically ill patient is best practiced in the safe confines of the intensive care unit (ICU). Transport of ventilated patients, however, remains a frequent challenge. Successful transport requires effective communication, appropriate planning, key personnel, and compact, rugged equipment. Clinicians should be aware of the physiologic effects of transport, frequency of adverse events, and methods to prevent complications.

Intrahospital Transport

Mechanically ventilated patients are moved frequently within the hospital. The most common destinations for patients transported from the ICU are computed tomography (CT) and the operating room with other radiologic modalities also being common.^{1–15} Most transports between the ICU and non–operating room destinations last 40 to 90 minutes.^{2,5–7,10,12–14} More recently, portable scanners have been used to perform CT scans in patients with traumatic brain injury to reduce the uncertainties of transport.

Magnetic resonance imaging is an increasingly common destination for the critically ill ventilated patient. Safe transport is more challenging in this setting because of limited patient access and the need for nonferrous equipment.

Interhospital Transport

Interfacility transport has increased in recent years owing to the regionalization of specialty care in neonatology, respiratory failure, trauma, transplantation, stroke, and cardiac disorders. The growth of hospital systems with acute and chronic care facilities represents another reason for interhospital transport as patients travel back and forth based on acuity. The current military operations in Iraq and Afghanistan have generated thousands of long distance interhospital transports requiring mechanical ventilation.

Interhospital transport can be accomplished using ground or air transport. Ground transport is the most readily available and least expensive, while also the least influenced by weather. There are few restrictions on weight and patient access can be quite good. Helicopters offer a significant improvement in speed but are expensive to operate. Patient care is made more difficult by weight limitations and a cramped, noisy cabin. Fixed-wing transport is the only viable option when critical patients must be moved over long distances.

Preparation and Planning

The American College of Critical Care Medicine suggests that all hospitals have a formalized plan addressing pretransport coordination and communication, composition of transport team, transport equipment, monitoring during transport, and documentation.¹⁶ Many adverse events associated with transport can be avoided with preparation and

communication.^{14,15,17–21} Patients having to wait at their destination can be particularly problematic and can be avoided with proper coordination.⁸

An experienced critical care nurse and respiratory therapist should accompany all mechanically ventilated patients during transport.^{16,22} The need for physician presence has been evaluated in pediatric transports but has not been clearly elucidated.^{23–25} The American College of Critical Care Medicine recommends a physician competent in airway management, advanced cardiac life support, and critical care medicine accompany all unstable patients.¹⁶ The American Association for Respiratory Care does not opine on physician presence, but does suggest that team members be skilled in airway equipment operation and troubleshooting.²²

Risks and Benefits of Transport

Considerable effort has been expended to catalog the risks of transport.^{3–8,10–15,17,18,26,27} The costs associated with transport include transport equipment, personnel, and managing complications of transport. Benefits may include discovering new pathology that changes treatment, intervention to address disease processes, or, in the case of interhospital transport, receiving care not available at the sending facility. Several investigations have noted that in two-thirds of transports to radiology, the patient's treatment course is unaltered.^{5,6} Head CT is least likely to result in a change in therapy, whereas abdominal CT is most likely to result in new findings and guide intervention.

Physiologic Effects and Complications of Transport

Transporting ventilated patients requires transitioning to portable equipment, repositioning the patient, and transfer from the ICU bed. Patients are often transported in the supine position, which changes respiratory mechanics and may alter hemodynamics. Many intrahospital destinations are distant from the ICU, located in facilities never designed to house critically ill patients. Both remote locations and poor physical plants may contribute to complications and untoward outcomes. Complications of intrahospital transport are frequently recalled as “the catastrophe in radiology.” During interhospital transport, patients must be cared for in the suboptimal confines of a transport vehicle. Transport complications range from minor changes in heart rate to cardiac arrest. [Table 27-1](#) lists common complications. The true complication rate is difficult to ascertain because of inconsistent definitions in existing studies.

Table 27-1: Commonly Reported Complications during Transport of the Mechanically Ventilated Patient**Cardiovascular**

Arrhythmia
 Hypotension
 Hypertension
 Tachycardia
 Bradycardia
 Myocardial ischemia
 Worsening heart failure
 Cardiac arrest

Respiratory

Hypoventilation
 Hyperventilation
 Hypoxemia
 Barotrauma

Neurologic

Elevated intracranial pressure
 Increased anxiety

Other

Increased risk of ventilator-associated pneumonia
 Bleeding or hemorrhage
 Hypothermia
 Effects of altitude on physiology^a
 Hypoxic hypoxemia
 Expansion of trapped gases (endotracheal tube, pneumothorax, ears, gastrointestinal tract)

Equipment Failure or Mishaps

Airway obstruction
 Extubation
 Intubation of right mainstem bronchus
 Gastric aspiration around endotracheal tube cuff
 Loss of battery power to monitoring equipment or ventilator
 Damage to equipment as a result of mishandling or falls
 Loss of oxygen supply
 Failure to reproduce ICU ventilation parameters with a portable ventilator or manual ventilator (loss of positive end-expiratory pressure, failure to trigger, inappropriate tidal volume or frequency)
 Effects of altitude on equipment performance^a
 Loss of venous access/indwelling vascular catheters
 Interruption of medications—continuous and intermittent
 Inadequate chest tube drainage

^aUnique to air transport.

Cardiovascular events are the most frequent complication during intrahospital and interhospital transport, and are reported in up to 50% of cases.^{4–7,11,13,14,17,28–33} An increase in heart rate is seen frequently as a result of anxiety, pain, and activity. Arrhythmias are common during the transport of high-risk cardiac patients, a finding complicated by the fact that routine lead II monitoring may be unable to detect early changes.^{34,35} Acute respiratory alkalosis resulting from hyperventilation alters myocardial irritability, leading to dysrhythmias.^{4,36–39} This finding commonly is associated with unmonitored manual ventilation but can occur with a ventilator when minute ventilation is unreliable.³⁹

Hypotension during transport may result from loss of intravenous access, interruption of vasoactive agents, pneumothorax, or bleeding. Alkalemia associated with hyperventilation likewise can cause hypotension.^{4,39,40} Excessive hyperventilation is associated with increased intrathoracic pressure, reduced venous return, and reduced cardiac output. During cardiopulmonary resuscitation, excessive ventilation reduces the efficiency of cardiopulmonary resuscitation and is referred to as “death by ventilation.” Hypertension can result from stress and anxiety, as well as from patient movement that encounters “bumps” in the transport path and jostling of patients that causes pain.^{3,5,6,41} Cardiac arrest has been reported during transport, but patient movement per se has not been directly implicated.^{17,31,32}

Respiratory Complications

Hypoxemia is a well-documented transport complication.^{2,4–6,14,30,32,33,40,42,43} It may result from loss of positive end-expiratory pressure (PEEP), changes in patient position, impaired secretion removal, equipment malfunction, and failure to reproduce ventilator settings adequately. During transport of adult patients, high inspired oxygen concentrations (FI_{O_2}) are used commonly as a matter of convenience.^{18,44} Elevated FI_{O_2} , however, may mask deterioration in lung function and contribute to absorption atelectasis. Patients requiring PEEP of more than 10 cm H_2O are vulnerable to deterioration in oxygenation during transport that lasts up to 24 hours.³

Hyperventilation is frequent with manual ventilation^{4,39,42,45} and with poor control of minute ventilation by portable ventilators.⁴² Sudden respiratory alkalosis can result in changes in cardiovascular function. Hyperventilation increases intrathoracic pressure, produces air trapping, reduces cardiac output, shifts the oxyhemoglobin dissociation curve to the left (hindering oxygen unloading), and causes cerebral and myocardial vasoconstriction. These combined effects may adversely affect patient outcome.⁴⁶

Hypoventilation is reported less frequently and generally is better tolerated.^{3–6,8,10–13,26,27} Decreased tidal volume contributes to atelectasis and acute respiratory acidosis, which may compromise cardiac function. Sedation and neuromuscular blocking agents may also contribute to hypoventilation.

Unplanned extubation is an infrequent but potentially catastrophic complication.^{17,30} Inability to establish an adequate airway is associated with hypoxemia and poor outcome in head-injured patients.^{19,47} Unintubated patients with marginal airway control may benefit from intubation before transport. Radiologic procedures requiring the patient to remain supine may compromise the tenuous airway.

Other Complications

Increases in intracranial pressure have been reported during transport and are associated with supine positioning, changes in ventilation, airway compromise, and hypoxemia.^{3,19,47}

Hypothermia during transport occurs because hallways, examination areas, and transport vehicles have imprecise environmental controls.¹⁰ Exposure of skin surfaces for adequate radiologic examinations and use of skin preparations for procedures further contribute to temperature loss. Blood loss has been reported during movement of patients with unstable fractures.¹⁰ Transport from the ICU is an independent predictor for the development of pneumonia, although a causative role cannot be definitively established.^{9,48} [Table 27-2](#) lists studies evaluating complications of transport.

Table 27-2: Comparison of Studies Evaluating Complications and Mishaps Intrahospital Transport of Critically Ill Patients

		Complications					
Year, Author	Complication Rate	Cardiovascular	Respiratory	Equipment	Other	Type of Ventilation	Attendants
1975, Waddell	7.2%	7.2% Hypotension, hypertension, tachycardia	0%	0%	Bleeding 1.6%	NR	RN, MD
1986, Insel	24%	24% Hypotension, arrhythmia	0%	0%	0%	Manual	RN, MD
1987, Braman	66%	25% Hypotension, arrhythmia	56% Hypocarbica, hypercarbia	5.5% Ventilator battery failure, Disconnected oxygen supply	0%	Manual $n = 20$ Ventilator $n = 16$	RN, MD, RRT
1988, Indeck	68%	61% Tachycardia, hypotension, hypertension	37% Hypoxemia, tachypnea	0%	0%	Ventilator	RN, MD RRT
1989, Weg	15%	0%	10% Hypoxemia, hypocarbica	5% Disconnected oxygen supply	0%	Manual	RN, RRT
1990, Andrews	51%	9% Hypotension 23% Rise in ICP 14% Hypertension	9% Hypoxemia	NR	0%	Ventilator	NR
1990, Smith	34%	NR	NR	34% ECG lead disconnected, monitor battery failure, ventilator disconnection	NR	NR	RN, MD, RRT

		Complications					
Year, Author	Complication Rate	Cardiovascular	Respiratory	Equipment	Other	Type of Ventilation	Attendants
1992, Hurst	66%	27% Tachycardia 36% Hypotension, hypertension	20% Tachypnea 2% Hypoxemia	5% Pulse oximeter failure, monitor battery failure	0%	Ventilator	RN, MD, RRT
1995, Szem	5.9%	1% Hypotension 1.5% Cardiac arrest	4% Hypoxemia	NR	NR	Manual or ventilator	MD, RRT
1995, Evans	53%	5.5% Arrhythmia 25% Tachycardia 39% Hypotension	17% Hypoxemia	11%	0%	Ventilator	NR
1995, Wallen	76%	47% Tachycardia 21% Hypotension	29% Tachypnea 6% Hypoxemia	10% Monitoring battery failure, ventilator disconnection	10% Hypothermia	Manual	RN, MD
1998, Stearly	15.5%	NR	NR	NR	NR	NR	RN
2004, Beckman	31%	3% Hypotension 3% Cardiac arrest	11% Hypoxemia or hypoventilation	34% Including unavailable devices, battery exhaustion, ventilator failure, exhaustion of oxygen supply	NR	Ventilator	RN
2006, Maza	46%	17% Hypotension 8% Hypertension	29% Hypercarbia	NR	66% Agitation	Ventilator	MD, RRT, RN

		Complications					
Year, Author	Complication Rate	Cardiovascular	Respiratory	Equipment	Other	Type of Ventilation	Attendants
2006, Gilman	23%	1% Hypotension 4% Hypertension 0.3% Bradycardia 2% Tachycardia	0.3% Hypoxemia	9% Equipment “problems”	7% Hypothermia	Ventilator	RN
2009, Zuchelo	100% (112 events in 44 transports)	12% Tachycardia 31% Hypertension	28% Hypoxemia 9% Hypercarbia 17% Acidosis	14% Exhaustion of battery or oxygen supply 3% Oximeter failure	3% Agitation 2% Emesis	Ventilator, manual ventilation	MD, RRT, RN

Abbreviations: ECG, electrocardiograph; ICP, intracranial pressure; MD, physician; NR, not reported; RN, registered nurse; RRT, registered respiratory therapist.

Special Considerations for Aeromedical Transport

Air transport subjects patients to altitude-induced physiologic alterations. Fixed-wing aircraft commonly pressurize to a cabin altitude of 8000 feet (barometric pressure of 565 mm Hg). Rotor-wing aircraft do not have pressurized cabins, leaving the patients and crew exposed to ambient barometric pressure. Flight altitudes are typically much lower in fixed-wing flight, but may reach physiologic significance when flying over high terrain.

The most relevant alterations in the hypobaric environment are hypoxemic hypoxia and expansion of gases. A change in altitude from sea level to 8000 feet is associated with hypoxemia in patients with normal and abnormal pulmonary function.^{49,50} At similar altitude changes, gases expand in volume by 30%. This may cause injurious pressure increases in endotracheal tube cuffs during ascent.^{51–53} Inconsequential volumes of trapped gases at sea level may cause significant physiologic effects at altitude, most dramatically in the setting of untreated pneumothorax. Less severe, but important, effects for patient comfort include ear pain secondary to pressure changes and expansion of gas in the gastrointestinal tract.

Ventilator performance can be adversely affected by changes in barometric pressure. Pneumatically operated devices will have alterations in tidal volume, frequency, and inspiratory time as altitude increases. Even sophisticated transport ventilators may deliver excessive tidal volumes in hypobaric environments.⁵⁴ Some newer ventilators automatically compensate for hypobaric effects on gas density.

Contraindications to Transport

There are few absolute contraindications to transport. Transport of the ventilated patient is best considered as transferring the ICU with the patient, not transferring the patient from the ICU. Patients occasionally are deemed too sick to transport. The specific contraindications to transport are (a) inability to maintain acceptable hemodynamic status, (b) inability to establish an adequate airway, (c) inadequate personnel, (d) inability to maintain adequate gas exchange, and (e) inability to monitor patient status effectively. Rescue therapies for acute respiratory distress syndrome are a relative contraindication to transport. If transport is required for potentially lifesaving therapy, then it should be considered. Safe transport with aggressive rescue therapies, including inhaled [nitric oxide](#), prone ventilation, inhaled prostacyclin, and high-frequency percussive ventilation, has been described.^{55–59}

Equipment and Monitoring during Transport

Appropriate equipment and monitoring are important to maintaining homeostasis and ensuring safety ([Table 27-3](#)). Monitoring during transport should emulate the ICU. Minimum requirements include electrocardiographic monitoring of heart rhythm and rate, invasive or noninvasive blood pressure monitoring, and pulse oximetry. Continuous capnography greatly enhances safety by immediately detecting inadvertent extubation, circuit disconnection, or ventilator failure. Its importance cannot be overstated. Monitoring should adapt to the needs of the patient and include additional pressure monitoring if intracranial pressure or central venous pressure catheters are present. Battery-powered intravenous pumps for medication delivery also are required.

Table 27-3: Monitoring and Lifesupport Equipment for Transport of the Ventilator-Dependent Patient

Monitoring
◦ ECG monitoring (rate and rhythm)
◦ Arterial blood pressure monitoring (invasive or noninvasive)
◦ Pulse oximetry
◦ Additional pressure monitoring (e.g., pulmonary artery, intracranial)
◦ Stethoscope
◦ End-tidal CO ₂ monitor
◦ Portable spirometer (if not integral to ventilator)
Support equipment
◦ Portable ventilator capable of providing required mode, tidal volume, and PEEP
◦ Airway maintenance (should include a difficult airway kit)
◦ Manual resuscitator and mask
◦ Oxygen supplies (one or two cylinders)
◦ Drug box (emergency drugs, patient-specific drugs, sedatives, IV fluids)
◦ Infusion pumps (battery-operated)
◦ Defibrillator (optional)
◦ Portable suction (optional)

Equipment for transport should be rugged, lightweight, reliable, and operate from battery power for at least 1 hour. Required equipment for all transports includes a physiologic monitor; pulse oximeter; ventilator; manual resuscitator with a PEEP valve and mask; [oxygen](#) supply with sufficient reserve; stethoscope; and emergency airway-management equipment. Sedative, analgesic, and neuromuscular blocking agents, as well as resuscitation drugs and fluids should be readily available during transport. Some means to humidify inspired gases and suction the airway should be available for interhospital and long intrahospital transports. A spirometer to monitor tidal volume is useful with ventilators that fail to provide a reliable tidal volume

and during the use of pressure ventilation. Airway carbon dioxide monitoring can be helpful when sedation reduces minute ventilation.

A dedicated transport cart or trolley may simplify transport but is an inefficient use of resources.^{6,34,45,60,61} The advantage of the cart is immediate availability of required devices. The disadvantages are cumbersome size and cost. A dedicated trolley for transport perhaps is best suited for interhospital transport.⁵⁹

Equipment Malfunction and/or Mishaps

Mishaps range from benign to catastrophic. The frequency of mishaps varies widely with the definition. Using a broad definition of mishap, up to one-half of patients suffer a mishap during transport, most of which are minor and do not cause adverse effects.^{8,14,15,17} Common mishaps include loss of vascular access or intravascular monitoring devices, disconnection from oxygen supply, disconnection from the ventilator, and improper care of chest tubes.^{4,6,8,13–15,17,30,31,41}

Equipment failure remains a common mishap during transport, frequently attributed to poor planning and carelessness.^{13,41} Reported failures include depletion of battery power of ventilator or monitoring equipment, exhaustion of portable oxygen supplies, ventilator failure, and damage to devices resulting from falls to the floor.^{4–6,8,13–15,17,26,30,31} Exhaustion of a compressed oxygen supply during use of a pneumatically powered ventilator terminates ventilation and can be a life-threatening mishap. Table 27-2 provides a comparison of studies evaluating complications and mishaps in intrahospital transport of critically ill patients.

Rationale for Use of a Portable Ventilator

Manual ventilation is commonly used during transport. It is inexpensive, simple, and requires only human power. During manual ventilation, however, the volume, frequency, and pressure applied are unknown. Several investigators have noted hyperventilation and acute respiratory alkalosis during manual ventilation resulting in cardiovascular complications.^{34,39,40,42,43,45–47} Aggressive manual ventilation can result in excessive airway pressures, causing barotrauma or volutrauma and worsening air trapping.⁶²

Manual ventilation can be successful in the hands of a skilled clinician with additional volume-monitoring and pressure-monitoring capabilities.^{10,13} Monitoring end-tidal CO₂ also may prove helpful in preventing hyperventilation.⁴⁰ Simple reasoning, however, dictates that manual ventilation cannot replicate the tidal volume, frequency, FI_{O2}, PEEP, and mode of ventilation with the consistency and precision of a ventilator. Maintaining constant PEEP with a manual resuscitator is difficult and failure to do so can lead to hypoxemia.^{2,18,43} Finally, manual resuscitators cause excessive work of breathing when spontaneous ventilations are present.^{63,64}

Comparisons of manual ventilation with a transport ventilator uniformly support the use of a ventilator to preserve normal gas exchange.^{4,39,42} A mechanical ventilator also allows monitoring and alarms. Modern transport ventilators provide the most commonly used ventilator modes, flow triggering, FI_{O2} and constant and PEEP.

Portable ventilators are more expensive than manual resuscitators, and many require a high-pressure (50 pounds per square inch gauge [psig]) gas supply. Although manual ventilation of the postoperative patient with normal lungs during transport to the recovery room is safe, ventilation of the patient requiring PEEP, elevated FI_{O2}, and constant volume is best accomplished by a ventilator.

Transport Ventilators

Technically, any ventilator that operates from a battery and either an internal gas source or a compressed gas cylinder could be considered a transport ventilator. Because of the demands of patient transport, however, these simple criteria are inadequate. Using performance to discriminate transport ventilators yields three categories: automatic resuscitators, pneumatically powered transport ventilators, and sophisticated transport ventilators.

Automatic Resuscitator

Automatic resuscitators provide ventilation at a set pressure. The user has limited control over respiratory parameters, and some do not allow any control of respiratory rate or tidal volume. The ability to apply PEEP is limited or nonexistent. As a function of pressure cycling, the respiratory rate and tidal volume vary with lung compliance. When compliance is low, tidal volume is small and frequency rapid; when compliance is high, tidal volume is high and frequency slow. These devices are powered pneumatically (requiring no electricity), and have only a mechanical, audible high-pressure alarm, and disposable pressure manometers. Spontaneous ventilation requires entrainment of ambient air, which reduces FI_{O_2} . Spontaneous respirations will cause dyssynchrony and increased work of breathing because of limited inspiratory flow rate and difficulty triggering. Automatic resuscitators are designed for use in the prehospital setting by personnel with limited expertise in mechanical ventilation.

The Vortran and Oxylator EM-100 are inexpensive examples of automatic resuscitators (Fig. 27-1). The Vortran has proven unreliable when the orientation is changed, failing to cycle.^{65,66} In the presence of a leak, if the set peak pressure cannot be reached, both devices will remain stuck in inspiration. Despite these limitations, successful use of the Vortran in intubated, closely monitored subjects has been reported.⁶⁷ Our experience with the Vortran, however, suggests it is unsuitable for transport of patients with even minor lung disease.

Figure 27-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Examples of automatic resuscitators, the Vortran (left) and Oxylator (right).

Simple Transport Ventilator

The simplest transport ventilators provide mechanical ventilation at a specified rate and volume and include a pressure-relief valve with an audible mechanical alarm. Most are powered and controlled pneumatically. In some instances, a battery allows for simple low-pressure and high-pressure alarms, as well as monitoring and display of airway pressure. Simple transport ventilators also are used primarily in prehospital settings by personnel with some training in mechanical ventilation. More complex than automatic resuscitators, they offer respiratory rate and tidal volume adjustment and can be used with spontaneously breathing patients.

Examples of Simple Transport Ventilators

The AutoVent 2000 (Fig. 27-2) and 3000 are pneumatically powered transport ventilators operating in the intermittent mandatory ventilation mode. FI_{O_2} is 100% and cannot be adjusted, and PEEP is not available. Mandatory breaths are time-triggered, flow-limited or pressure-limited, and time-cycled, while spontaneous breaths are pressure-triggered, pressure-limited, and pressure-cycled. During spontaneous breaths, the patient breathes from the demand valve at 48 L/min. If patient demand exceeds 48 L/min, ambient air is drawn into the valve, diluting the FI_{O_2} . There is no monitoring. An audible alarm sounds if airway pressure exceeds 50 cm H_2O .

Figure 27-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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The AutoVent 2000 simple transport ventilator.

The AutoVent 2000 is designed for adults and allows control of breath rate and tidal volume. The AutoVent 3000 is designed for adults and children and allows the user to control inspiratory time. The AutoVent 4000 offers additional alarms, manometer, adjustable airway pressure relief valve, and continuous positive airway pressure (CPAP) mode; it allows selection of FI_{O_2} 0.65 or 1. The ability of the AutoVent 4000 to provide a consistent FI_{O_2} is altered by reduced compliance. As the lung becomes stiffer, the venturi mechanism reduces air entrainment and FI_{O_2} increases while tidal volume falls.

The AutoVent ventilators consume approximately 0.5 L/min of gas to operate the logic. The higher the frequency setting, the higher is the gas consumption. The inspiratory flow is fixed at 48 L/min. In patients with an active respiratory drive, the flow capabilities of the AutoVent devices may result in flow starvation, increased work of breathing, and asynchrony.

The Uni-Vent 706 (Fig. 27-3) is a pneumatically powered, electronically controlled ventilator delivering an FI_{O_2} of 1. Controls set inspiratory flow and a series of rate, inspiratory time, and inspiratory-to-expiratory timing (I:E) ratio combinations appropriate for adult and pediatric ventilation. Inspiratory flow is limited from 0 to 90 L/min, and a high-pressure limit can be set (60 or 80 cm

H₂O) at the patient valve, which activates an audible alarm when exceeded. Prehospital use of the Uni-Vent 706 by paramedics during cardiopulmonary resuscitation of intubated victims has proven successful.⁶⁸

Figure 27-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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The Uni-Vent 706 simple transport ventilator.

Sophisticated Transport Ventilator

A sophisticated transport ventilator is capable of performance comparable to an ICU ventilator. These devices may have built-in compressors or turbines to generate positive pressure without compressed gas and contain an air-oxygen blender. Most offer extensive control of ventilator parameters and comprehensive alarms. Sophisticated transport ventilators are intended for interhospital or intrahospital transport of critically ill patients. These devices should be capable of ventilating the sickest patients.

Examples of Sophisticated Transport Ventilators

The IC-2A is a flow controller that can be triggered by pressure or time or manually; it is also pressure- or flow-limited and time-cycled (Fig. 27-4). It requires a compressed-gas source for delivery to the patient, as well as for the fluidic logic circuit. The IC-2A delivers an FI_{O_2} of 1. Controls include the mode of ventilation, inspiratory and expiratory time, inspiratory flow, sensitivity, and PEEP/CPAP level. Airway pressure is displayed on an aneroid pressure gauge. The IC-2A is often used for ventilation in the magnetic resonance imaging scanner because it has no ferrous components. The excessive gas consumption should be considered when gas supplies are limited. The IC-2A operation during synchronized intermittent mandatory ventilation limits the flow delivered to the set flow on the ventilator, potentially increasing the work of breathing. Our experience with this ventilator suggests most patients have to be sedated to tolerate the operational limitations.

Figure 27-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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The MRI compatible IC-2A.

The LTV 1200 offers the performance of a critical care ventilator at a size and weight appropriate for a transport ventilator (Fig. 27-5). It provides synchronized intermittent mandatory ventilation, controlled mechanical ventilation, pressure support, and CPAP modes. Pressure-controlled and volume-controlled breaths are possible. The integral turbine permits mechanical ventilation without a compressed gas source. **Oxygen** can be supplied from a low-flow or compressed-gas source. The user has extensive control of ventilator parameters. Airway pressure, total breath rate, exhaled tidal volume, total minute volume, I:E ratio, calculated peak flow, and patient effort are monitored and displayed. The bias flow during exhalation (10 L/min) increases gas consumption and should be considered when planning transports. The LTV 1200 offers additional features, including ventilator presets, and an **oxygen** conservation feature to decrease gas consumption during transport. The LTV 1200 also utilizes an electronic control of PEEP, compared to the a spring-loaded valve used with the LTV 1000. This allows the LTV 1200 to apply pressure breaths relative to baseline pressure.

Figure 27-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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 The LTV 1200 sophisticated transport ventilator.

The Uni-Vent 754 is an electrically powered flow or pressure controller offering controlled mechanical ventilation, synchronized intermittent mandatory ventilation, and CPAP modes of ventilation. The internal compressor allows use without a compressed gas source. The operator can set mode of ventilation, breath rate, inspiratory time, tidal volume, sensitivity, PEEP, FI_{O_2} , and peak inspiratory pressure limit. Airway pressure is displayed, as well as the airway pressure waveform, ventilator breath rate, inspiratory time, I:E ratio, tidal volume, FI_{O_2} , mean airway pressure, and baseline airway pressure. A full set of alarms is present. Inspiratory flow is limited to 60 L/min, which can cause dyssynchrony in spontaneously breathing patients. The inability to provide pressure-limited breaths is a minor limitation. The Uni-Vent 754 has a proven track record in transporting ventilated military casualties by U.S. Air Force critical care, air transport teams.

The Uni-Vent 731 transport ventilator (Fig. 27-6) is the next generation of Impact ventilators and is an electrically powered device that offers multiple improvements over the Uni-Vent 754. Most notable is the ability to provide pressure-limited breaths, integral pulse oximeter, and a 10-hour battery life. The peak flow in volume ventilation has also been increased to 100 L/min. The 731 has the capability to provide closed-loop control of inspired oxygen concentration using the pulse oximeter input. The target arterial oxyhemoglobin saturation (SpO_2) can be set between 92% and 99%. If SpO_2 falls below 88%, the FI_{O_2} is increased rapidly to alleviate hypoxemia. This system has potential important advantages during transport when physiologic changes occur quickly and access to the patient may be limited. At the time of writing, this system is not available in the United States.

Figure 27-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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 The Impact 731 sophisticated transport ventilator.

Attributes Common to All Transport Ventilators

Weight

These devices must be person-portable and able to be mounted on a variety of platforms. In our mind, portable means that a respiratory therapist can carry the device to and from destinations. In our opinion, ICU ventilators that can be mounted on the ICU bed are in reality not portable. A maximum weight of 8 kg facilitates these requirements.

Durability

These devices should be compact, simple to operate, durable, and unaffected by extremes of heat, cold, or vibration. Proper shielding is required to limit emission of unacceptable levels of electromagnetic energy. Automatic resuscitators in particular should function properly after prolonged storage with minimal maintenance.

Power Requirements

Automatic resuscitators are pneumatically powered, negating the need for batteries. Simple and sophisticated transport ventilators often are powered electrically and pneumatically.

Transport ventilators use a variety of battery supplies. Sealed lead-acid batteries have a low energy density but are durable and inexpensive. Sealed lead-acid batteries hold a charge even when stored for long periods of time. Nickel-cadmium (NiCad) batteries have a higher energy density but are more expensive. Nickel-metal hydride batteries have even a higher energy density than NiCad batteries and are more expensive. Lithium-ion batteries have the highest energy density and are the most expensive. NiCad, and to a lesser extent nickel-metal hydride, batteries (but not sealed lead-acid or lithium-ion batteries) can suffer from voltage memory and should be stored in the discharged state. Both lithium-ion and NiCad batteries will self-discharge during storage. Duration of ventilator operation on battery is significantly impacted by ventilator settings in all devices.

Oxygen consumption should be less than 5 L/min, keeping in mind that oxygen consumption varies with ventilation mode and the use of PEEP. The device should have the capability of generating flow without a compressed-gas source and offer a means of enriching the inspired-gas mixture with oxygen from a low-pressure source such as a concentrator.

Controls

Controls should be large and not easily adjusted accidentally. Any display should be viewable from an angle and in variable ambient light.

Safety

Automatic resuscitators and simple transport ventilators have basic safety features, whereas sophisticated transport ventilators possess safety features comparable with those of critical care ventilators. A pressure-relief valve that vents gas to the atmosphere at a preselected peak inspiratory pressure is essential. Violation of the high-pressure limit should be signaled by a visual or audible alarm. An antiasphyxia valve that allows the patient to breathe from ambient air in the event of power source failure is desirable.⁶⁹ Battery-powered ventilators should be equipped with a “low battery” alarm as the battery nears depletion. If the compressed-gas source falls below operating pressure of the ventilator or is empty or disconnected, an audible or visual alarm should sound. Visual alarms are critical when ventilators are used in environments with high levels of ambient noise such as aircraft.

Assembly and Disassembly

These devices should be designed so that incorrect circuit installation is impossible. Typically, there is a single-limb circuit with an external expiratory valve. Patient valves located at the endotracheal tube should be cleared of secretions easily.

Performance Issues Relevant to All Types of Transport Ventilators

Delivered Tidal Volume

Several investigators have demonstrated unreliable tidal volume delivery with transport ventilators, encouraging monitoring with a portable spirometer.^{54,70–76} This includes low tidal volumes in the face of low lung compliance and excessive tidal volume under normal loads and at altitude.

Imposed Work of Breathing

Transport and home-care ventilators often used for transport have been shown to have an unacceptable imposed work of breathing.^{74–76} Recent studies of new devices show a wide range of capabilities among transport ventilators.^{70,76,77} Transport ventilators with flow triggering and PEEP compensation consistently offer the least imposed work of breathing.

Battery Life

Exhaustion of equipment batteries, including transport ventilators, has been reported in multiple studies of critical care transport.^{8,14,15,17,30} The consequences of ventilator battery exhaustion can be catastrophic during long transport of patients requiring advanced mechanical ventilation. The battery life among common transport ventilators varies greatly and may differ from values in the user's manual.^{78–80} Use of PEEP, pressure-control ventilation, and increasing FI_{O_2} will shorten battery life in electrically powered ventilators while having little effect on pneumatically driven models.^{79,80} Battery life should be considered when planning transport. Provisions should be made for extra batteries or use of AC power if the transport duration exceeds battery life.

Oxygen Consumption

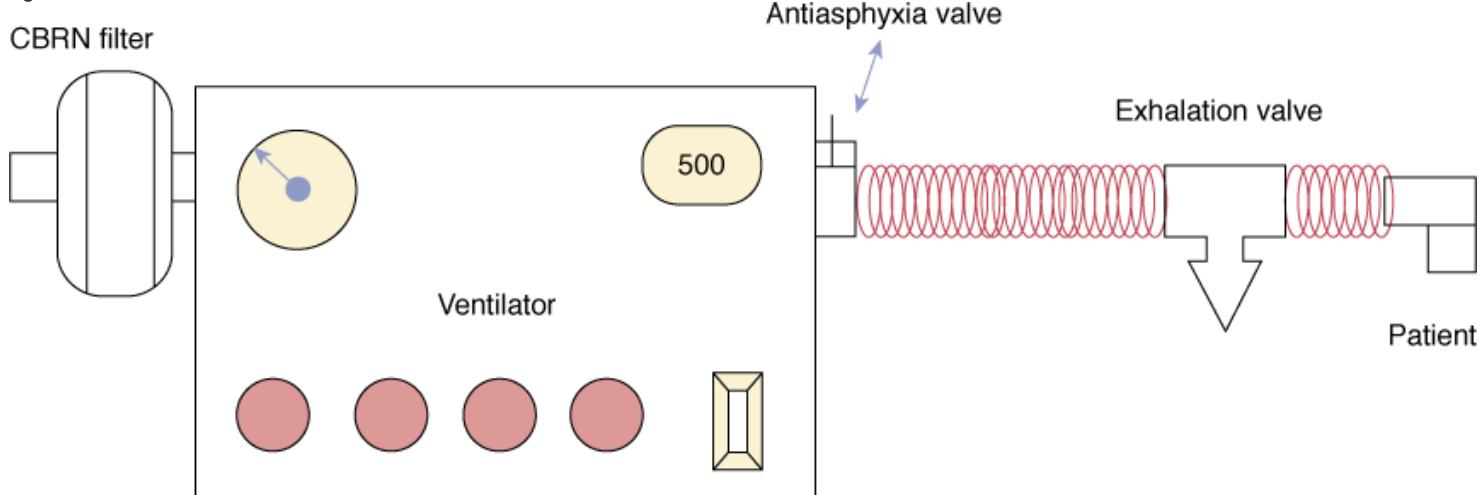
It is critical to estimate expected oxygen consumption to plan for a safe transport. It is intuitive that high FI_{O_2} will increase oxygen consumption, but accounting for bias flow, leaks, changes in patient condition, and variability between ventilators is more difficult. A study of military patients found oxygen consumption was 1.6 to 10.2 L/min using the Uni-Vent 754, with a broad range of mechanical ventilation parameters.⁸⁰ The LTV 1000 and LTV 1200 offer a cylinder-duration calculator that slightly underestimates cylinder duration.⁸¹ The Dräger portable ventilators and Impact 731 have a continuous display of oxygen consumption on the front panel display. Continuous assessment of oxygen consumption will allow early identification of consumption exceeding estimates, which should prompt a search for leaks, increase efforts to conserve oxygen, and take steps to terminate or divert the transport as needed.

Operation in a Hazardous Environment

Worldwide concern over terrorist events and natural disasters has led manufacturers and investigators to consider use of portable ventilators in hazardous environments. This includes following radiation exposure and airborne biologic hazards (anthrax, botulism, nerve agents). A chemical-biologic-radiologic-nuclear filter can be placed on the inlet of the compressor of the ventilator to protect the gas delivered to the patient. A recent study demonstrates that most portable ventilators fail to isolate the patient from ambient air during normal operation.⁷⁷ In ICU ventilators, the antiasphyxia valve is kept closed by an electric circuit. The antiasphyxia valve is only opened in the instance of ventilator failure. Portable ventilators typically use a one-way leaf valve in the inspiratory limb between the ventilator outlet and the patient (Fig. 27-7). If patient demand exceeds

ventilator output under normal conditions, gas can enter the circuit. This not only exposes the patient to any ambient agents, but reduces FI_{O_2} . Only the Impact 731 routes all gas, including gas via the antiasphyxia valve, through the chemical-biologic-radiologic-nuclear filter. Breathing through the antiasphyxia valve prevents contamination of inspired gas, but does so at the cost of increased work of breathing for the patient secondary to increased resistance.

Figure 27-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Placement of the chemical-biologic-radiologic-nuclear (CBRN) filter on the ventilator and position of the antiasphyxia valve. Under normal conditions, if patient demand exceeds ventilator output (flow), room air is entrained, which allows contamination of the patient gas and reduces FI_{O_2} .

Ancillary Equipment for Transport

Oxygen Supply

All ventilators require an oxygen supply source. Both compressed-gas cylinders and liquid systems can fulfill this need. Intrahospital transport usually is accomplished with an E cylinder or two E cylinders yoked together. These provide 630 and 1260 liters of gas, respectively. An H cylinder contains 6900 liters and may be required for longer transports. The H cylinder is 152 cm in height, weighs 68 kg, and requires its own attendant.

Liquid-oxygen systems can provide 860 cubic feet of gaseous oxygen for every liquid cubic foot. Most liquid systems, however, cannot operate at 50 psig.

Oxygen concentrators designed for home and ambulatory use have recently been investigated for use during mechanical ventilation.⁸² Their utility has not been fully defined, but the concept is theoretically very promising.

Humidification

A passive humidification device, or “artificial nose,” is ideal for transport. Use of an artificial nose may result in a progressive increase in breathing circuit resistance, and the patient should be monitored for signs of expiratory-flow restriction. Dead space should be accounted for during low-volume ventilation. Premoistening an artificial nose is inadvisable, does not improve efficiency, and only serves to further increase flow resistance.

Summary

Safe transport of the mechanically ventilated patient requires effective communication, appropriate planning, the presence of key personnel, and compact, rugged equipment. Equipment should meet the demands of patient acuity, and personnel should have the requisite skills and training for the task at hand. Clinicians should be aware of the most frequent complications of transport along with methods of prevention and treatment. The nuances of individual ventilators from battery life and [oxygen](#) consumption to imposed work of breathing must be appreciated.

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